REQUIREMENT ANALYSIS:

1. Introduction

- **Purpose:** To define the functional and non-functional requirements for the Smart Library System.
- **Scope:** Includes the request portal, approval workflows, inventory management, and notification system.

2. Functional Requirements (The "Must-Dos")

These describe the specific behaviors of the system.

ID	Requirement Name	Description
FR-01	**Request Submission**	Users must be able to submit a "Borrow Request" via a Service Portal form.
FR-02	**Automated Validation**	The system must check if the `u_book` record is currently "Available" before triggering the flow.
FR-03	**Approval Routing**	Requests must be routed to the **Librarian** role for manual oversight.
FR-04	**Status Updates**	The system must automatically update the Record Status to "Borrowed" on approval and "Rejected" on denial.
FR-05	**Due Date Logic**	The system must calculate a return date (e.g., $Today + 14\ days$) upon successful checkout.
FR-06	**Notifications**	The system must send automated emails to the requester when an approval decision is made.

3. Non-Functional Requirements (The "Quality" Constraints)

These describe how the system should perform.

- **Security:** Only users with the `librarian` role should be able to approve or reject requests.
- **Usability:** The "Borrow Request" form should take a user less than 60 seconds to complete.
- **Reliability:** The Flow Designer logic must handle errors (Error Handler) without getting stuck in a "Waiting" state indefinitely.
- **Scalability:** The system should support up to 500 concurrent book records without performance lag.

4. Data Requirements

This defines the structure of your ServiceNow tables.

Table: Book [u_book]

- **Title:** String
- **Author:** String
- **ISBN:** String
- **Status:** Choice (Available, Borrowed, In Repair)

Table: Borrow Request [u_borrow_request]

- **Requester:** Reference (sys_user)
- **Book:** Reference (u_book)
- **Approval State:** Choice (Requested, Approved, Rejected)
- **Due Date:** Date/Time

5. System Workflows (Logic Flow)

This section documents the logic you showed in your screenshot:

1. **Trigger:** Record created in `u_borrow_request`.
2. **Action:** Ask for Approval (Approver = Librarian).
3. **Decision:** * If **Approved**: Update `u_book` status to "Borrowed" and notify user.
    - If **Rejected**: Update `u_borrow_request` status to "Rejected" and notify user.

6. User Personas

- **Student/Patron:** Can create requests and view their own borrowing history.
- **Librarian:** Can view all requests, approve/reject them, and manage the book inventory.
- **System Admin:** Manages the ServiceNow flow, table permissions, and email templates.